

Research Method of Data Science

Week 7: Quantitative Research Design

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Week 7: Quantitative research design

Tuesday: making a question measurable and a design credible

Thursday: testing and improving your own research design

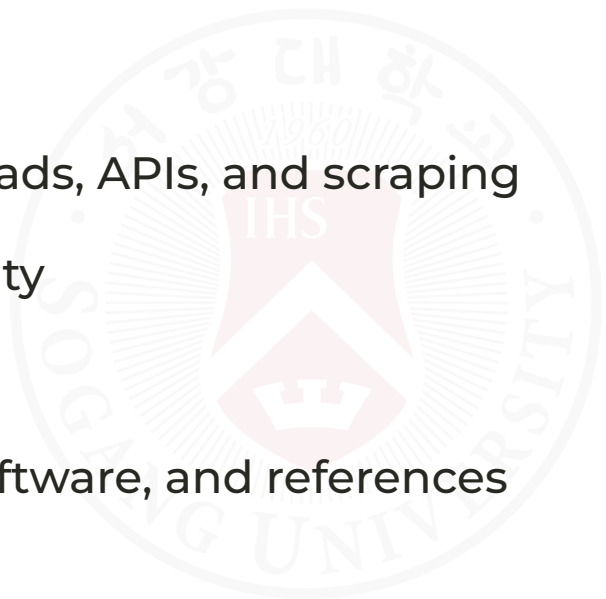
From a research question to evidence you can defend.



Agenda

Tuesday — Week 7, Meeting 1

1. From concepts to measurable variables
 2. Surveys, sampling, and selection
 3. Data collection: sources, downloads, APIs, and scraping
 4. Missingness and threats to validity
 5. Preparation for the design clinic
- Appendix: templates, examples, software, and references

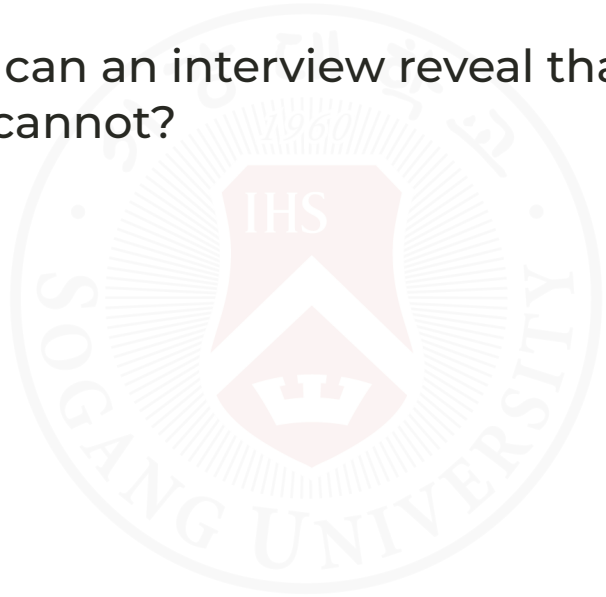


From Week 6 to Week 7

Week 6: How do students describe and interpret their AI use?

Week 7: How is a defined measure of AI use associated with a defined measure of learning?

Pair discussion — 2 minutes: What can an interview reveal that a test score cannot? What can a test score reveal that an interview cannot?



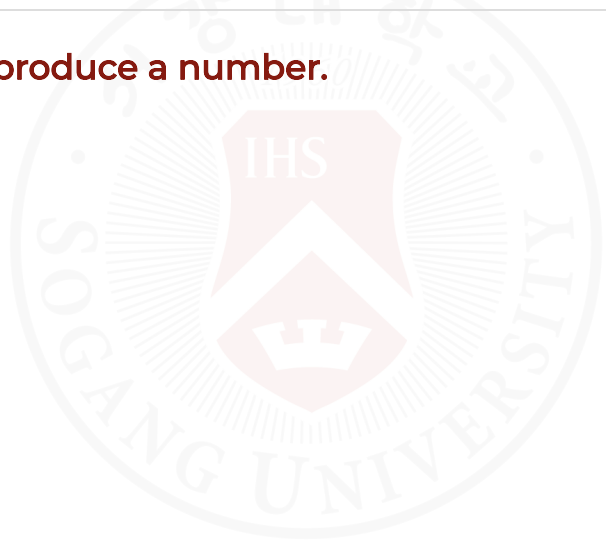
The seal of Chung-Ang University is a circular emblem. It features a central shield with a crown on top and the letters 'IHS' inside. Above the shield is the year '1960'. The shield is surrounded by a circular border containing the university's name in Korean and English.

1. From concepts to measurable variables

Start with the claim you want to make

Purpose	Example question	Design implication
Describe	How often do students use AI?	Define population, period, and use
Explain an association	Is AI reliance related to unaided performance?	Measure both; consider alternative explanations
Estimate a causal effect	Does access to an AI tutor change later performance?	Establish a credible comparison

Your method must support the claim, not just produce a number.



Concept → operational definition → variable

Concept: a theoretical idea, such as AI reliance.

Operational definition: the rule used to observe it.

Variable: the recorded value for each unit.

Example: number of study tasks in the past seven days completed using AI-generated answers without checking them.

That variable captures **one dimension** of reliance—not the entire concept.

Make the measurement decisions explicit

Decision	Illustrative choice
Unit of analysis	Individual university student
Population/context	Students in a specified institution and semester
AI measure	Share of study tasks using unverified AI answers
Learning measure	Score on a later test completed without AI
Time order	Use recorded before the later test
Other relevant variables	Prior achievement, study time, course difficulty

Availability is still to be checked.



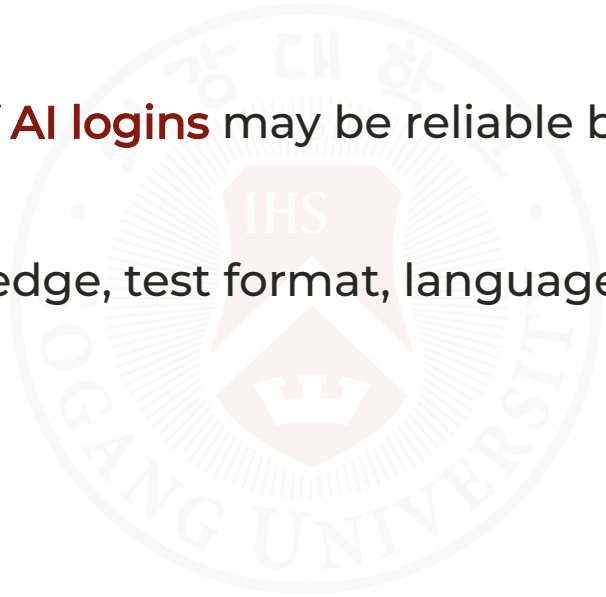
Reliability and validity are different

Reliability: consistency of measurement under relevant conditions.

Construct validity: whether evidence supports interpreting the measure as the intended concept.

A consistently recorded **number of AI logins** may be reliable but a weak measure of **dependence on AI**.

A test score can reflect task knowledge, test format, language, or motivation—not only critical thinking.



Measurement repair – pairs, 5 minutes

A student proposes:

| “I will measure critical thinking by asking: How good are you at critical thinking? (1–5)”

1. Identify **two weaknesses**.
2. Suggest a more defensible measurement strategy.
3. Name one limitation that would remain.

Prepare a revised operational definition.

| Hint: consider what the measure actually captures, how it is scored, and what it can and cannot support.

The scale affects what you can do

Example	Measurement level / caution
Main study tool	Categories; numeric labels are not quantities
One agreement item: strongly disagree → strongly agree	Ordered categories; spacing is not guaranteed
Number of study sessions	Count; define what counts as a session
Income / housing price	Specify currency, period, inflation adjustment

Do not treat **missing**, **not applicable**, and **zero** as the same value.

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2. Surveys, sampling, and selection

A survey is a measurement instrument

Before opening **Google Forms or Qualtrics**, decide:

- What concept does each item measure?
- Can respondents know and recall the answer?
- Are the period and response options clear?
- Are consent, access, and privacy arrangements appropriate?

Pilot: ask a few suitable people how they interpreted the items.

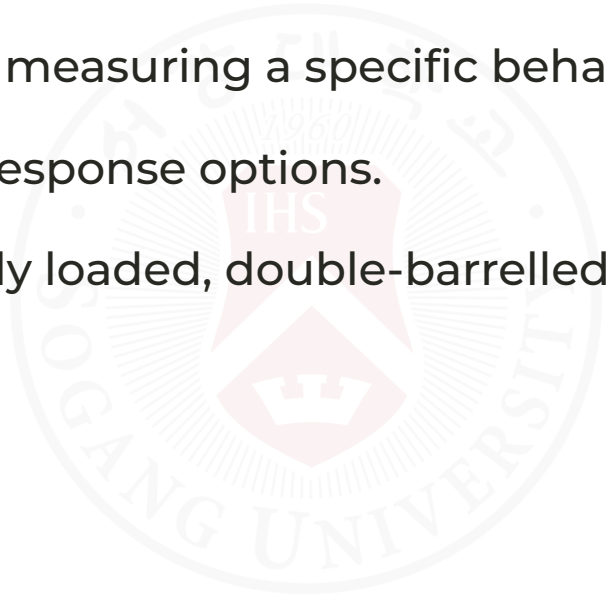
This helps identify any ambiguities or misinterpretations before the full survey is distributed.

Repair the survey item – pairs, 4 minutes

| “Don’t you agree that excessive AI use harms learning and makes students lazy?”

1. Identify the wording problems.
2. Replace it with **one neutral item** measuring a specific behaviour.
3. Add a time period and suitable response options.

| Hint: avoid leading, emotionally loaded, double-barrelled, undefined excessive, and implied causality.



Who is included at each stage?

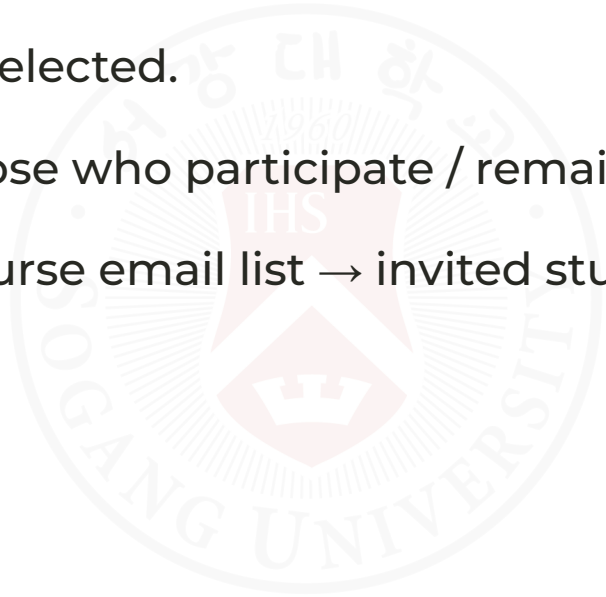
Target population: everyone your claim concerns.

Sampling frame: the list or mechanism through which they can be reached.

Selected sample: those invited or selected.

Respondents / analytic sample: those who participate / remain in the analysis.

Example: university students → course email list → invited students → completed responses.



Sampling: choose for your purpose

Strategy	What it involves	Main concern
Simple random	Chance selection from a defined frame	Frame coverage and nonresponse
Stratified random	Random selection within defined groups	Group sizes and weighting
Cluster	Select groups, then all/some members	Dependence within groups
Convenience / volunteer	Accessible people or self-selected participants	Selection and generalization

A large convenience sample is still a convenience sample.



Random selection $\neq$ random assignment

Random selection: who enters the sample?

→ supports population inference, subject to coverage and response.

Random assignment: who receives an intervention?

→ supports a causal comparison under relevant assumptions.

You can have either, both, or neither.

Example: A study might randomly select participants from a population (random selection) and then randomly assign them to different treatment groups (random assignment).

Selection challenge – 4 minutes

A link is posted in an AI enthusiasts' group. **800 students** respond.

The researcher concludes: “University students frequently rely on AI.”

In pairs:

1. Who is the target population? Who can enter the sample?
2. Does 800 solve the selection problem?
3. Rewrite the claim and suggest one design improvement.

| Hint: consider the frame, recruitment, and nonresponse.

3. Data collection: sources and access methods

Primary and secondary: origin and purpose

Primary data: collected specifically for your study.

Examples: your approved survey, experiment, or observation protocol.

Secondary data: existing records reused for your question.

Examples: official statistics, an existing survey, administrative records.

API and scraping describe how you access data—not automatically whether the study uses primary or secondary evidence.

API = application programming interface, a documented way for software to request information. Scraping = automated extraction of permitted page content.

Both can be used for primary or secondary data collection.

Data structure changes the possible question

Structure	Example unit / time	Can support...
Cross-section	Many students measured once	Differences at a given period
Time series	Vietnam measured annually	Changes over time in one unit
Panel	Provinces measured repeatedly	Within-unit and between-unit patterns

2015–2025 inclusive gives only 11 annual observations for one country.



Choose the simplest adequate access route

Route	When useful	Tools
Download CSV / Excel	Provider supplies a documented table	Browser, Excel, pandas
API	Structured, repeatable requests are supported	Colab, Python requests
HTML extraction	Needed information is in permitted page content	pandas.read_html, BeautifulSoup
Manual coding	Small set needs interpretation	Excel, Word, coding log

A downloaded table can be more suitable than a complex scraper.



An API request has a research specification

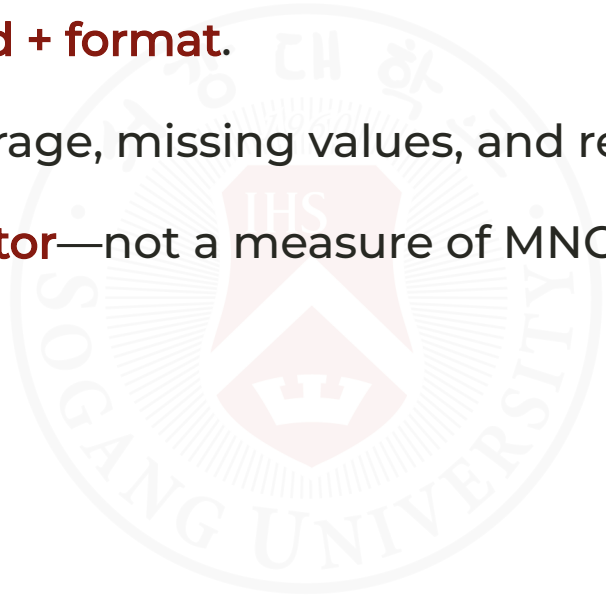
World Bank example: **Vietnam population, 2015–2025**

<https://api.worldbank.org/v2/country/VNM/indicator/SP.POP.TOTL?date=2015:2025&format=json>

Specify **country + indicator + period + format**.

Then check units, definitions, coverage, missing values, and revisions.

Population is a possible **denominator**—not a measure of MNC activity.



Web scraping: extraction is only the beginning

Web scraping: using software to extract information from web pages.

Example: collect advertised rent, area, and listing date.

But ask:

- Are these unique properties or duplicate advertisements?
- Are prices asking prices or completed transactions?
- Which neighbourhoods, platforms, and dates are covered?
- What information is absent or changes over time?

Scraping demonstration: read a small table

```
from io import StringIO
import pandas as pd

html = """<table>
<tr><th>id</th><th>rent_krw</th></tr>
<tr><td>A01</td><td>800000</td></tr>
<tr><td>A02</td><td>-</td></tr>
<tr><td>A01</td><td>800000</td></tr>
</table>"""
df = pd.read_html(StringIO(html), na_values=["-"])[0]
```

Predict: How many rows? How many unique IDs? What needs checking?



Choose tools—and document access

Beautiful Soup: inspect and extract elements from HTML.

pandas: import tables and inspect records.

For any automated collection:

- Check provider terms, permissions, and published access guidance.
- Prefer documented downloads/APIs; respect request limits.
- Do not bypass logins or access restrictions.
- Avoid collecting identifiable personal data for this lab.

Save the source, retrieval date, query, and collection code.

Data-source audit – pairs, 3 minutes

For the rent table, decide:

1. What is one row supposed to represent?
2. What metadata are missing?
3. Can it answer “What rent do all city residents pay?”

One sentence: “This source can tell us , *but cannot yet tell us* .”

| Hint: consider unit, time, geography, definition, and uniqueness.

4. Missingness and threats to validity



Missingness is information about the process

Distinguish:

- **Unit nonresponse:** an invited person provides no response.
- **Item nonresponse:** a participant skips a question.
- **Coverage / collection gaps:** some groups, years, or pages are absent.

A blank may mean unknown, withheld, not applicable, or extraction failure.

First investigate why; then choose and justify a treatment.

Three missing-data ideas

Mechanism	Illustrative situation
MCAR: missing completely at random	Random technical loss unrelated to measured/unmeasured values
MAR: missing at random, conditional on observed data	Missing scores depend on recorded attendance, not on scores after conditioning
MNAR: missing not at random	Low scorers are less likely to provide scores even after accounting for observed variables

These are assumptions about the process, not labels proved by a blank cell.

What could make your conclusion misleading?

Threat	Design question
Measurement / construct validity	Does the variable represent the concept?
Selection	Who is excluded or more likely to enter?
Confounding	What influences both exposure and outcome?
Reverse causality	Could the proposed outcome influence exposure?
Time/history	What else changed during the period?
External validity	To whom and where does the finding apply?

| Consider the context and generalizability of your findings.

Before-after $\neq$ the effect of a policy

Illustrative design: Seoul housing prices before and after a tax change.

Prices may also change with interest rates, credit conditions, supply, and expectations.

A stronger design needs a defensible comparison and clear timing.

Safe question for a limited design: “How did observed prices change during this period?”

| This question focuses on the observed change without making causal claims.

Group-level relationships need group-level claims

Example: provincial immigrant share and reported crime rate.

Check crime definitions, population denominators, reporting, tourism, and time coverage.

An association between provinces does **not** establish that immigrants commit more crime—or that inflows caused a change.

Ecological inference problem: group patterns do not identify individual behaviour.

| Do not infer individual-level conclusions from group-level data.

Validity repair – pairs, 5 minutes

Fictional study: 40 volunteers report weekly AI use and their own grades. The researcher writes: “AI improves achievement.”

1. Identify **three specific threats**.
2. Propose **two feasible design improvements**.
3. Rewrite the conclusion within the evidence boundary.

| Hint: consider selection, measurement, confounding, and time order.

Repair the design before adding complexity

For each threat, write:

Threat → mechanism → response → remaining limitation

Example:

Volunteer bias → enthusiasts respond more often → recruit through a broader frame and report response → nonresponse may still differ.

No software can recover information that the design never observes.

5. Preparation for the design clinic



Bring evidence for your individual project

For Thursday, bring:

1. Your current research question.
2. Your literature matrix and **two verified relevant sources**.
3. One candidate dataset/source and its documentation, if found.
4. A draft list of concepts and possible measures.

If a source is unavailable, record **what is missing**.

For qualitative projects: examine a quantitative alternative without being required to adopt it.

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Thursday – Individual research design clinic

Agenda

Thursday — Week 7, Meeting 2

1. Build your question–evidence alignment
2. Audit sources and collection feasibility
3. Peer critique and individual revision
4. Midterm review
5. Submit your design-clinic note

Today you improve the design for your **complete individual research project**.



What you will produce today

A short **design-clinic note** containing:

- Question and literature link
- Unit, population/context, and period
- Concepts, variables, and operational definitions
- Source, collection route, and sample/coverage
- Method, main threats, and defensible claim boundary
- Changes made after feedback and next action

Use the appendix template. **This is a working design note, not a new full proposal.**

Individual alignment draft – 12 minutes

Use your own question and two verified sources.

1. State the claim purpose, unit, context, and period.
2. Define your main concepts and proposed measures.
3. Explain what each source contributes to your design.
4. Identify the evidence and comparison your answer needs.

Underline one decision you cannot yet justify.

Source audit – 12 minutes

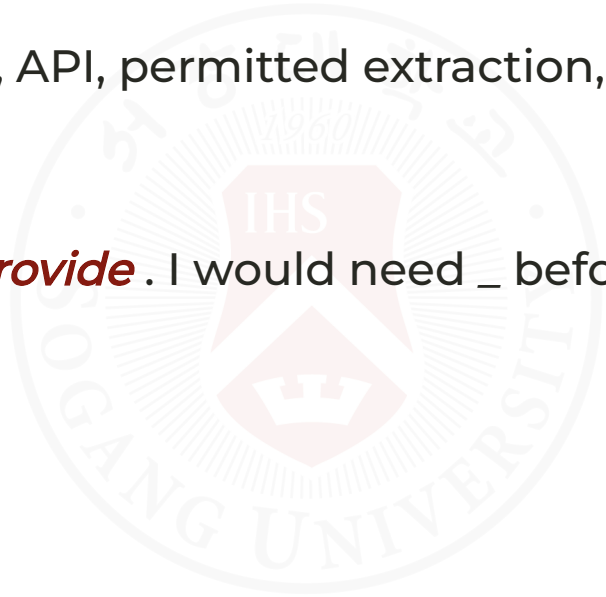
Audit your source or use the fictional cards in the appendix.

Check **origin, variables, unit, coverage, time, access, missingness, and documentation.**

Choose an access route: download, API, permitted extraction, or planned primary collection.

Write:

| “This source fits ***; it does not provide***. I would need _ before analysis.”



Peer design clinic – 12 minutes

Six pairs; six minutes per student.

- **2 minutes:** explain your alignment and one uncertainty.
- **3 minutes:** partner checks the design using the checklist.
- **1 minute:** agree on the highest-priority revision.

Partner feedback:

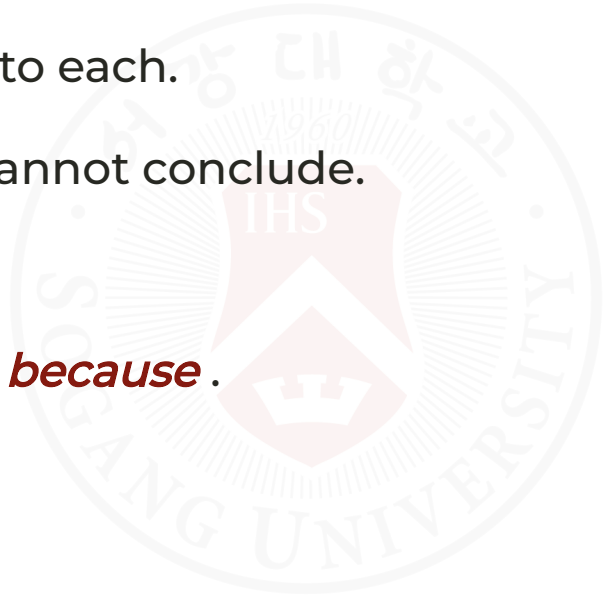
One coherent link · one mismatch · one feasible repair

Individual revision – 12 minutes

Revise your own note using the feedback.

1. Repair the most important mismatch.
2. Add **two threats** and a response to each.
3. State what your study can and cannot conclude.
4. Write your next feasible action.

Keep a short revision log: changed *because*.



Midterm review – what to practise

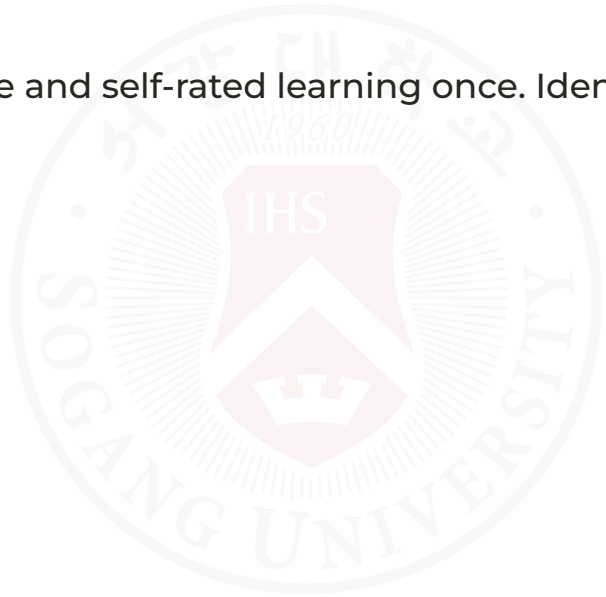
Practise explaining and evaluating:

- Research questions, problems, and contribution
- Literature synthesis and verified sources
- Qualitative coding, interpretation, and evidence boundaries
- Measurement, sampling, data-source fit, and validity
- Transparent workflows and verification

Official scope, format, permitted tools, and AI rules follow the instructor's assessment instructions.

Midterm practice – 8 minutes individually

- 1. Literature:** Fictional Source A finds better assisted task scores; Source B finds no difference on a later unaided test. Write a synthesis without treating the results as identical outcomes.
- 2. Qualitative:** “AI helps me begin, but I attempt the problem myself before asking.” Suggest a code and a bounded interpretation.
- 3. Design:** A survey of one class measures AI use and self-rated learning once. Identify one measurement issue, one selection issue, and a defensible claim.



Submit your design-clinic note – 7 minutes

Check that your note includes:

- ✓ Question–literature–data–variable–method alignment
- ✓ Source and access plan; sampling/coverage boundaries
- ✓ Two threats and responses; limits of the claim
- ✓ Peer-feedback revision and next action

Submit individually through **Cyber Campus**, following the posted instructions.

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Appendix – clinic materials and references

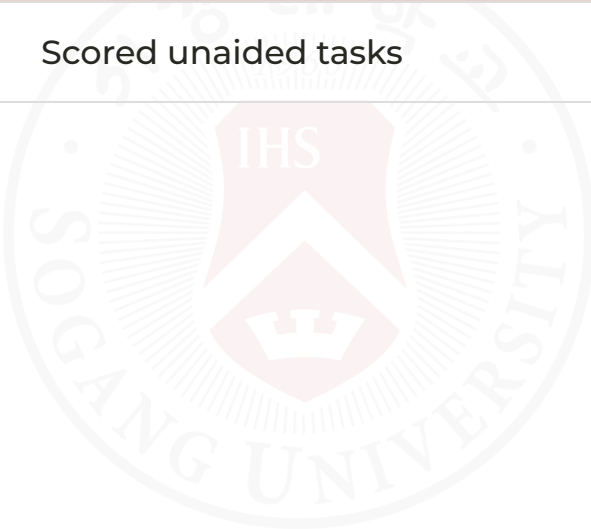
Design-clinic template

Field	Your decision and justification
Question / claim purpose	What exactly will you answer?
Literature link	Which verified sources support definitions/design?
Unit / population / period	Who or what; where; when?
Concepts / measures	Operational definition and limits
Source / access / sample	Where evidence comes from; who is included
Method / comparison	How evidence will answer the question
Threats / responses	Two mechanisms; repairs; residual limitations
Claim boundary / next step	What you cannot infer; immediate action

Operationalization worksheet

Concept	Definition from literature	Observable rule	Unit / period	Limitation
AI use frequency	How often AI is used for studying	Days used in past 7 days (0–7)	Student/week	Recall; not reliance
Reliance	Using AI in place of own effort	Requires a justified instrument or observable protocol	Specify	Not established by login count
Learning	Performance on specified learning objectives	Scored unaided tasks	Student/test	Task and scoring dependence

Examples are illustrative; adopt measures only after justification.



Fictional source cards: AI and learning

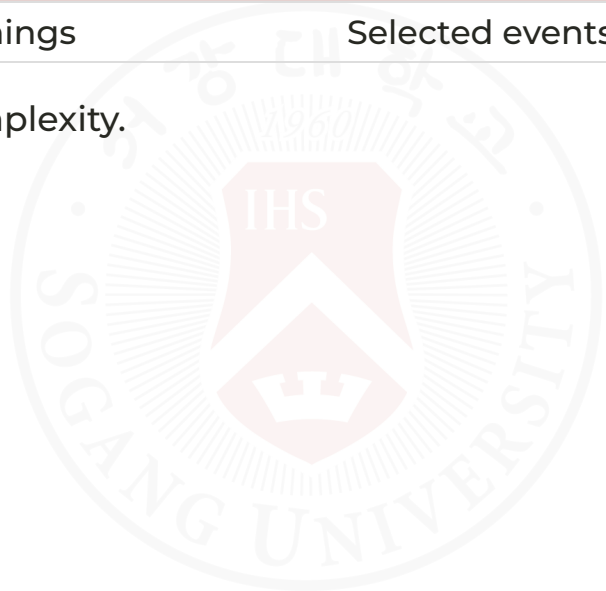
Card	Evidence available	Limitations
A: volunteer survey	80 students, one course; weekly use; self-rated learning	No objective test; self-selection; one time point
B: study records	120 consenting students; baseline and later unaided scores; AI-use log	Participation/access uncertain; log not reliance; no random assignment
C: public aggregate	Annual national internet-use share	Wrong unit, population and construct for student learning

Which fits an **association** question best? Which cannot answer it?

Fictional source cards: MNCs and Vietnam

Card	Evidence available	Fit issue
D	Annual total FDI inflows, exports, GDP growth: 2015–2025	11 years; FDI not identical to MNC manufacturing
E	Provincial manufacturing investment and output: five years	Potential panel; definitions, overlap and access need checking
F	News articles about selected factory openings	Selected events; not a complete investment series

Narrow the claim before increasing model complexity.

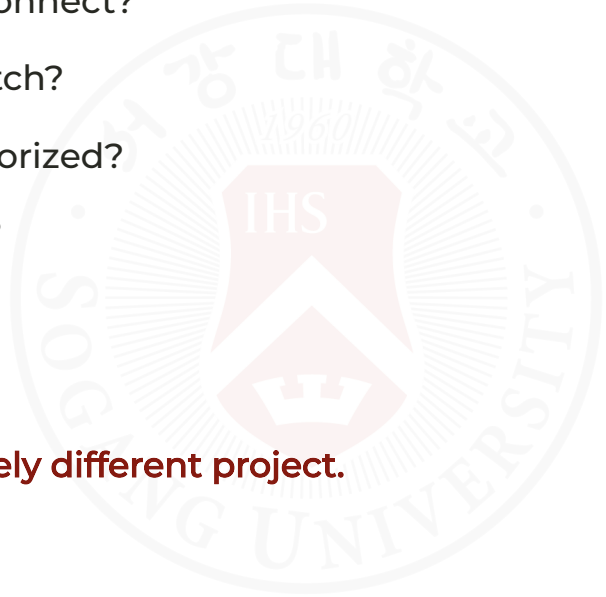


Peer clinic checklist

Ask your partner:

1. Does the question specify the intended claim?
2. Do the literature, concepts, and measures connect?
3. Do source units, population, and period match?
4. Is collection feasible and appropriately authorized?
5. Who/what could be systematically missing?
6. Does the method support the claim?
7. Is the main limitation explicit?

Recommend one priority repair, not a completely different project.



A minimum data-source log

Record:

- Provider, original dataset title, source URL, retrieval date
- Definition of one row; geography; period; coverage
- Variable definitions, units, missing codes, revisions
- Access route, query/filter, file version, usage conditions
- Transformations and checks; storage location

Keep raw files unchanged; document derived files separately.



Software: what each tool contributes

Task	Accessible option	Research responsibility
Instrument draft	Word / Google Docs	Wording and measurement
Survey delivery	Forms / Qualtrics, subject to access	Sampling, consent, storage
Metadata / dictionary	Excel / Sheets	Definitions and missing codes
Data access / inspection	Colab + pandas / requests	Source fit and reproducibility
HTML parsing	Beautiful Soup / pandas	Coverage and extraction checks
Literature links	Zotero + your matrix	Verify original sources

Advanced scraping frameworks are optional, not required.

Other sampling approaches

Approach	Example	Caution
Systematic	Every kth record after a random start	Periodicity; frame order
Quota	Recruit until subgroup targets are met	Usually nonprobability within groups
Snowball	Participants refer others	Network dependence and exclusions
Purposive	Select information-rich cases	Explain selection and inferential purpose

Choose according to the question; explain the consequences for inference.

Validity: a final reference map

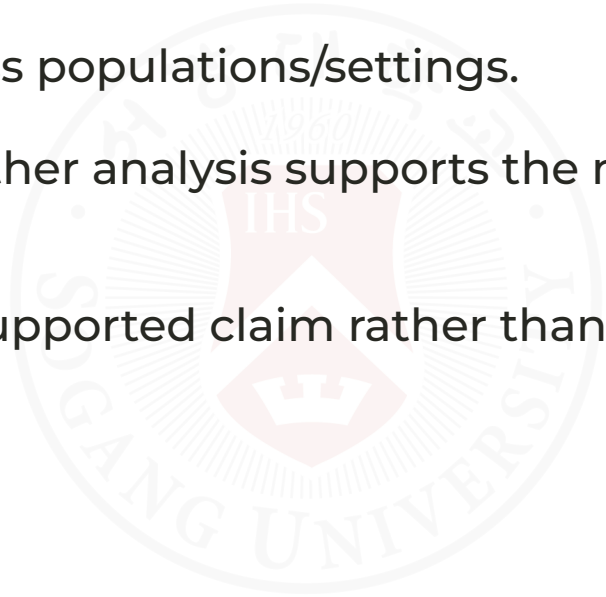
Construct validity: meaning of the measures.

Internal validity: credibility of a causal interpretation.

External validity: applicability across populations/settings.

Statistical conclusion validity: whether analysis supports the numerical conclusion, given precision and assumptions.

A good project makes a modest, supported claim rather than a broad, fragile one.



Verified methods and software references

- [Pew Research Center: writing survey questions](#)
- [AAPOR: best practices for survey research](#)
- [World Bank: Indicators API documentation](#)
- [pandas: read_html documentation](#)
- [Beautiful Soup: documentation](#)

Consult original source documentation before adopting data or code.



Example API URL and Python follow-up

Vietnam population, 2015–2025: World Bank API

Optional later exploration:

```
import requests
url = "https://api.worldbank.org/v2/country/VNM/indicator/SP.POP.TOTL"
r = requests.get(url, params={"date": "2015:2025",
                             "format": "json"}, timeout=30)
r.raise_for_status()
payload = r.json()
# Inspect metadata, records, missing values and pagination.
```

Network-dependent illustration; not required for the clinic.

